day18\1_reduceFunction.js

```javascript
//syntax of reduce function:
//1. const res = ar.reduce(callBack Function, initialization)
//2. const res = ar.reduce((accumulator, currentValue) => {
// }, initialization)
//initialization: it is the starting value of accumulator.
//currentValue: this is a current value in the array
const ar = [10, 20, 30];
const res1 = ar.reduce((accumulator, currentValue) => {
    accumulator = accumulator + currentValue;
    return accumulator;  //accumulator: which accumulates the values
}, 0);
console.log(res1);


//another way: using compact arrow function
const res2 = ar.reduce((accumulator, currentValue) => accumulator + currentValue, 0);
console.log(res2);


//code snippet
//In JavaScript, hasOwnProperty() returns true only if the property exists directly on the object itself, not somewhere up its prototype chain.
const aray = ["orange", "apple", "banana", "apple", "orange", "banana", "orange", "apple", "grapes", "apple"];
const answer = aray.reduce((accumulator, currentValue) => {
    if(accumulator.hasOwnProperty(currentValue))
    {
        accumulator[currentValue]++;
    }
    else{
        accumulator[currentValue] = 1;
    }
    return accumulator;
}, {}); //an empty object is passed in the initialization
console.log(answer);

//another way to write the above code snippet using conditional operator:
const ans5 = aray.reduce((accumulator, currentValue) => {
    accumulator.hasOwnProperty(currentValue) ? accumulator[currentValue]++ : accumulator[currentValue]=1;
    return accumulator;
}, {})
console.log(ans5);


//code snippet
const ans6 = aray.reduce((accumulator, currentValue) => {
    accumulator.hasOwnProperty(currentValue) ? accumulator[currentValue]++ : accumulator[currentValue]=1;
    return accumulator;
}, {orange: 3, apple: 4, grapes: 107}) //here, the values are also initialized into the empty object.
```

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console.log(ans6);



//code snippet
const ar5 = [1, 2, 3, 4];
const ans = ar5.reduce((acc, curr) => {
  acc.push(curr * 2); // modify and add each item
  return acc;
}, []);  //an empty array is passed in the initialization
console.log(ans);



//hasownproperty working:
let obj = {
    a: 1,
    b: 2
}
console.log(obj.hasOwnProperty("a")); //yes, key = a is present in the object, so true is in the output.
console.log(obj.hasOwnProperty("c")); //key = c is not there in the object, therefor the output is false.
```